Ayyagari Krishna Vaibhav

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Professional Summary

Driven 3rd-year Integrated M.Tech (CSE) student with a strong foundation in software engineering, systems programming, and full-stack development. Experienced in leading technical projects and developing impactful applications, including geospatial analysis systems, enterprise web apps, and system metrics tools. Eager to leverage skills in Python, GIS, and software architecture to contribute to ISRO-NRSC's data processing and remote sensing initiatives.

Education

University of Hyderabad (UoH) — Hyderabad, India
Integrated M.Tech in Computer Science Engineering (CSE) — 3rd Year

Technical Skills

- **Languages:** C, Python, Java, JavaScript, TypeScript, PHP
- **Web & Frameworks:** React, HTML, CSS
- **Domain Skills:** Remote Sensing Data Processing (Landsat/Sentinel), GIS Tools, Systems Programming, Software Architecture, Web Scraping

Professional Experience

Shatpadh Technologies Private Limited — Hyderabad, India
Director & Co-Founder — Jan 2022 – 2025

- Leading the end-to-end development of comprehensive software solutions and managing technical project lifecycles.
- Architecting robust, scalable applications to solve complex operational challenges, overseeing both front-end user experience and back-end logic.

Major Projects

Thermal Infrared Urban Heat Island Mapping and Alert System

- **Objective:** To map, analyze, and monitor urban temperature variations using satellite data to identify heat islands and generate automated environmental alerts.
- **Details:** Processed satellite imagery (Landsat/Sentinel) and applied GIS mapping techniques using Python to visualize thermal data. Designed the alert mechanism to flag abnormal temperature spikes in dense urban zones.

Placement Management System (UoH x SCIS)

- **Objective:** To streamline the university recruitment lifecycle for students, faculty, and hiring administrators.
- **Details:** Built a secure, full-stack platform managing student profiles, job postings, and interview schedules, ensuring robust data integrity and a seamless user interface.

SCIS Online Peer Review System

- **Objective:** To digitize and facilitate a seamless academic peer-review process for research and coursework.
- **Details:** Developed an interactive web application allowing for secure document submission, blinded reviews, and automated status tracking.

UoH Alumni Web Scraper

- **Objective:** To automate the extraction and aggregation of alumni professional data for university networking and record-keeping.
- **Details:** Engineered a robust scraping script handling pagination and data parsing, converting raw web data into a structured, searchable format.

Electricity Bill Generator Application

- **Objective:** To calculate and automate the generation of utility billing statements based on dynamic usage logic.
- **Details:** Implemented core computational logic to process varied tariff slabs, taxes, and user data, outputting accurate and formatted billing records.

Academic & Independent Projects

Software Metrics Tool for C Language — *Under Faculty Guidance, UoH*

- Developed an independent utility tool to analyze code complexity and efficiency metrics specifically for the C programming language.
- Built scripts for automated metric extraction and evaluated low-level memory allocation patterns to help improve system maintainability.

Open Source Contributions

Jenkins

- Successfully contributed to the Jenkins open-source automation server by submitting and merging pull requests, demonstrating an ability to navigate and improve large, complex legacy codebases.